# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: INTERNET PROGRAMMING**
**COURSE CODE: CSA43**

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**COURSE OUTCOME COVERED**

**CO-1:** Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

| | |
|---|---|
| Assessment Tool: | Code along exercise: Make your first HTML page together with CSS |

**Weightage: 10%**

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**QUESTIONS:** **Total Marks: 10**

## Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

## Data Snippets Provided

**Snippet 1: Video Embedding Options**

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

**Snippet 2: Form Submission Methods**

```
A. GET
B. POST
C. PUT
D. DELETE
```

**Snippet 3: Navigation Elements**

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

**Snippet 4: XML Syntax**

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

**Snippet 5: Web Communication Protocols**

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

## Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

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***Code of Conduct:***

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Case | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25 | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts |
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning |
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided |
| Clarity | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand |

# CSA43 – INTERNET PROGRAMMING

## ASSESSMENT TOOL 3: WEB DEVELOPMENT STANDARDS IMPLEMENTATION

*Detailed Solutions and Technical Justifications*

Course Outcome CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery.

Case-study approach: Each answer identifies the correct option, explains the relevant concept, applies it to the startup scenario, and analyses the alternatives.

## 1. Correct HTML Tag for Video Embedding

**Correct option:** B — <video src="video.mp4" controls></video>

The standard HTML5 <video> element embeds a video file in a web page. The src attribute gives the location of the video resource, while controls asks the browser to display its built-in playback controls, such as play, pause, volume, and fullscreen. This makes it appropriate for the startup's homepage promotional video.

Example implementation:

```
<section>\n  <h2>Watch Our Promotional Video</h2>\n  <video src="video.mp4" controls
width="600">\n    Your browser does not support HTML video.\n  </video>\n</section>
```

The fallback text appears if the browser cannot render the video element. In a production website, the video file must exist at the specified path, and additional formats or captions may be provided when needed.

Analysis of the alternatives:

A. <media src="video.mp4"></media> — Incorrect because <media> is not a standard HTML element for embedding video.

B. <video src="video.mp4" controls></video> — Correct because <video> is the HTML element designed for video content.

C. <movie src="video.mp4"></movie> — Incorrect because <movie> is not a standard HTML element.

D. <vid src="video.mp4"></vid> — Incorrect because <vid> is not a standard HTML element.

Conclusion: Option B follows HTML standards and provides native browser playback controls.

## 2. Appropriate HTTP Method for Registration Form Submission

**Correct option:** B — POST

The POST method is appropriate for a registration form because the browser sends the form fields in the HTTP request body for server-side processing. Unlike a typical GET request, POST does not append the submitted form values to the URL query string. A registration operation may create a new user account, so POST is a suitable method.

Example implementation:

```
<form action="/register" method="POST">\n  <label for="username">Username:</label>\n  <input
id="username" name="username" required>\n\n  <label for="email">Email:</label>\n  <input
type="email" id="email" name="email" required>\n\n  <label for="password">Password:</label>\n
<input type="password" id="password" name="password" required>\n\n  <button
type="submit">Create Account</button>\n</form>
```

Why POST is appropriate:

1. Form values are sent in the request body rather than the URL query string.

2. It is suitable for submitting information that the server must validate and process.

3. It supports operations such as creating a new account, subject to the server's implementation.

Important security clarification: POST is not encryption. Sensitive registration data must be transmitted over HTTPS, which protects data in transit using TLS. The server should validate inputs, apply appropriate access controls, and store passwords using a secure password-hashing method.

Analysis of the alternatives:

A. GET — Usually used to retrieve a resource. Parameters are commonly visible in the URL and may be recorded in browser history or logs, so it is unsuitable for password submission.

B. POST — Correct for submitting registration details to the server for processing.

C. PUT — Commonly used to create or replace a resource at a known URI; it is not the usual choice for a conventional HTML registration form.

D. DELETE — Requests deletion of a resource and does not match the purpose of registering a user.

Conclusion: Option B is the best answer. HTTPS and server-side security controls are additionally required for secure registration.

## 3. Proper Semantic Element for Navigation

**Correct option:** B — <nav>

The <nav> element represents a section of a web page containing major navigation links. Semantic elements describe the purpose of content instead of merely grouping it. Using <nav> makes the page structure easier to understand for developers, browsers, and assistive technologies.

Example implementation:

```
<header>\n  <h1>Startup Website</h1>\n  <nav aria-label="Main navigation">\n    <a
href="index.html">Home</a>\n    <a href="about.html">About</a>\n    <a
href="services.html">Services</a>\n    <a href="contact.html">Contact</a>\n  </nav>\n</header>
```

The aria-label provides a descriptive name for the navigation landmark, which is useful if a page contains multiple navigation sections. The links should point to valid destinations.

Analysis of the alternatives:

A.  — A generic container with no inherent semantic meaning. It can group elements, but it does not specifically identify navigation.

B. <nav> — Correct because it explicitly identifies a navigation section.

C. <section> — Groups thematically related content, normally with a heading; it is not the most specific element for primary navigation.

D. <header> — Represents introductory content or a section header. It may contain navigation, but <nav> is the dedicated semantic element for navigation links.

Conclusion: Option B is the most semantically appropriate and accessible choice for the main navigation menu.

## 4. Correct XML Syntax and Errors in the Alternatives

**Correct option:** B — <note><to>User</to><from>Admin</from></note>

XML (Extensible Markup Language) represents structured data using tags. A well-formed XML document must have exactly one root element, each start tag must have a matching end tag, and elements must be properly nested. XML tag names are case-sensitive.

Correct XML example:

```
<?xml version="1.0" encoding="UTF-8"?>\n<note>\n  <to>User</to>\n  <from>Admin</from>\n</note>
```

In this example, <note> is the single root element, and <to> and <from> are its child elements. Each element is closed correctly and the nesting order is valid.

Analysis of the alternatives:

A. <note><to>User</to><from>Admin</from> — Incorrect because the root element <note> has no matching closing tag.

B. <note><to>User</to><from>Admin</from></note> — Correct because every element is closed and the elements are properly nested.

C. <note><to>User</from></note> — Incorrect because <to> is opened but closed with </from>. Opening and closing tag names must match.

D. <note to="User"></note — Incorrect because the final closing tag is missing the > character. As written, the markup is incomplete.

Application to the case: The startup should correct the XML configuration so it is well-formed before the application attempts to parse it. A parser can reject malformed XML, which may prevent the configuration from loading.

Conclusion: Option B is valid XML syntax among the given choices.

## 5. Correct Protocol for Web Communication

**Correct option:** B — HTTP

HTTP (Hypertext Transfer Protocol) is an application-layer protocol that supports request-response communication between a web client, such as a browser, and a web server. A request identifies a resource or asks the server to perform an operation; the response provides a status code and, where applicable, content such as HTML, images, or scripts.

Basic request-response sequence:

1. A user requests a web page through the browser.
2. The browser sends an HTTP request to the server.
3. The server processes the request and prepares a response.
4. The server returns a status code and the requested content, if applicable.
5. The browser interprets the response and renders the page.

Analysis of the alternatives:

A. FTP (File Transfer Protocol) — Primarily used to transfer files between systems.

B. HTTP (Hypertext Transfer Protocol) — Correct; it is used for web client-server communication.

C. SMTP (Simple Mail Transfer Protocol) — Used to send email messages.

D. SNMP (Simple Network Management Protocol) — Used to monitor and manage network devices.

Security note: HTTPS is HTTP protected by TLS. It helps provide confidentiality and integrity for data exchanged between the client and server. Modern web applications should use HTTPS rather than unencrypted HTTP.

Conclusion: Option B is the correct answer for the protocol listed in the assessment. HTTPS is the secure form used for production websites.

## Consolidated Answer Key

| Question | Correct option | Reason in brief |
|---|---|---|
| 1 | B — <video> | Standard HTML element for embedding video. |
| 2 | B — POST | Submits registration fields in the request body. |
| 3 | B — <nav> | Semantic element for major navigation links. |
| 4 | B — Valid XML | All elements are correctly closed and nested. |
| 5 | B — HTTP | Protocol for web client-server request-response communication. |

## Conclusion

The case study illustrates that standards-compliant web development requires the correct HTML element for each purpose, an appropriate HTTP method for each operation, well-formed XML syntax, and a suitable communication protocol. For this scenario, the answers are <video>, POST, <nav>, the correctly closed XML structure, and HTTP. Secure deployment additionally requires HTTPS and suitable server-side validation and data protection.